

From spreadsheet guesswork to governed answers

Lumora Clinical Research × Databricks · A governed third-party data
marketplace

The outcome: faster, complete, governed vendor decisions

~\$600K / year

Illustrative annual value:
analyst time, avoided duplicate
data buys, better renewals

Days → minutes

Vendor-sourcing requests
answered in the same session,
with the SQL shown

3 of 3 vendors

Complete, deterministic
answers – the old pattern
returned 1 of 3

Today: the vendor catalog lives in spreadsheets — and so do the risks

- Feasibility, RWE and commercial teams ask: which vendor covers this data type, in this country, for this therapeutic area?
- The catalog is a folder of SharePoint workbooks; a homegrown assistant pastes them into an LLM prompt
- Answers drift: the same question returns different — and incomplete — vendor lists
- Contract cost and vendor contact details are visible to anyone who can open the files
- No evaluation, no guardrails, no audit trail — and renewals slip past without a decision

For the executive sponsor (COO / Chief Data & Analytics Officer)

What you get

Lower 3PD spend: duplicate purchases and blind renewals surface early

Faster study start-up: feasibility questions answered in minutes

Lower data risk: cost and PII governed by policy, audited

KPIs you can track

3PD spend under management and % renewed with a decision

Request turnaround time (target: same day)

Access-policy coverage of sensitive fields (target: 100%)

For the domain owner (Head of Data Partnerships & RWD Sourcing)

What changes for your team

Self-service answers from the business, with the SQL shown

Data-quality rules written in plain English

One place to manage vendors, coverage, contracts

KPIs you own

Answer completeness and accuracy (eval-scored)

Catalog data-quality pass rate

Renewal pipeline: contracts expiring with spend at risk

One integrated journey on Databricks

1 · Lakeflow	2 · Unity Catalog	3 · Lakebase	4 · Gen AI	5 · Genie Agent	6 · Databricks App
Auto Loader pipeline ingests raw vendor catalogs → Bronze → Silver → Gold; DQ expectations	Lineage, comments, column masks & row filters, metric views	Millisecond serving of vendor tiles to the app	Agent Bricks supervisor + Knowledge Assistant; MLflow eval; AI Gateway guardrails	Natural language → SQL; every matching vendor, SQL shown	Marketplace UI with AI assistant, plus AI/BI dashboard with Ask Genie

Live demo: the journey a sourcing request takes

- App → "Which vendors provide HCP data in Egypt?" – all 3 vendors, with counts
- Same box → "How does AfriHealth derive its counts?" – answered from documents, cited
- Genie → expand the SQL; the same answer from a governed metric view
- Dashboard → filter HCP + Egypt; Ask Genie; spend tiles masked by role
- Guardrails → a hostile or PII-bearing prompt is blocked before the model
- MLflow → every answer traced and scored; then a DQ rule written in English

Trust by design: governed data, governed AI

Governed data

Contract cost visible to finance only;
contact email/phone masked

Commercial contacts row-filtered by role
Lineage and audit for every query

Governed AI

Safety + PII guardrails at the prompt
boundary

Eval gate caught a masked-cost leak in
documents before go-live

Rate limits and usage tracking cap spend

Value model (illustrative assumptions – replace with Lumora's numbers)

Lever	Assumption	Formula	Annual value
Analyst time on sourcing requests	600 requests/yr; 6.0 h → 0.5 h; \$95/h loaded	$600 \times 5.5 \text{ h} \times \95	\$313,500
Duplicate / overlapping data purchases avoided	\$3.6M 3PD spend (18 vendors × ~\$200K); 5% avoided	$\$3.6\text{M} \times 5\%$	\$180,000
Renewal optimization	30% of spend renews/yr; 10% negotiated saving	$\$3.6\text{M} \times 30\% \times 10\%$	\$108,000
Total illustrative value	Platform run cost assumed \$50K/yr (to be sized)	$\approx 12\times \text{value} / \text{cost}$; payback ≈ 1 month	\$601,500
Conservative case	Half the requests, half the time saved, half the spend levers	$\$78,375 + \$90,000 + \$54,000$	\$222,375 ($\approx 4.4\times$)

Proof points from the reference build

- Flagship question: 3 of 3 matching vendors, deterministic, SQL shown (old pattern: 1 of 3)
- Same answer from a governed metric view — one definition of "coverage"
- Non-finance users get cost masked; Genie explains "masked for your role"
- LLM-judge evaluation on 4 question classes: correctness, relevance and safety all 1.0
- Eval gate caught a governance leak (masked cost in documents) before go-live — fixed, re-indexed, re-run clean
- Guardrails: hateful prompt blocked; SSN and email anonymized before the model
- Plain-English DQ rule enforced in a Lakeflow pipeline: 28 of 126 coverage rows flagged (98 kept)
- One Lakeflow Job runs it end to end: 18 vendors, 126 coverage rows, 72 documents indexed

Rollout: value in 30 days, production in 90

Days 0-30

Land 3-5 real vendor catalogs

Genie on governed Gold

Baseline turnaround & completeness

Days 31-60

Masks, metric views, dashboard

Supervisor + documents

Eval gate on real questions

Days 61-90

App to sourcing teams

All vendors, scheduled ingest

Track spend & renewal KPIs

The ask

- Executive sponsor: approve a 30-day pilot and name a domain owner
- Domain owner: share 3–5 vendor catalogs and 20 real sourcing questions
- Together: confirm the value-model inputs and success KPIs
- Databricks: deploy the reference build in Lumora's workspace and size the run cost



databricks